

StaySense BI Strategy Note

Business Intelligence Strategy for an Airbnb Analytics Platform

Section 1- Context

StaySense BI is a Business Intelligence solution developed using the Airbnb NYC dataset to support data-driven decision-making for property managers and business analysts. The platform provides interactive dashboards and analytical insights to monitor listing performance, pricing trends, neighbourhood demand, and customer preferences, enabling informed strategic and operational decisions.

Section 2- Recommended KPIs

1. **Average Listing Price** – Tracks the average price of available listings to monitor pricing competitiveness.
2. **Occupancy Indicator** – Measures listing availability throughout the year to estimate occupancy trends and demand.
3. **Average Review Rating** – Evaluates customer satisfaction based on guest reviews.
4. **Revenue Potential** – Estimates earning potential using listing price and availability information.
5. **Listings by Neighbourhood** – Identifies areas with high property concentration for market analysis.
6. **Room Type Distribution** – Monitors the popularity of entire homes, private rooms, and shared rooms.
7. **Host Performance** – Tracks the number of listings managed by each host to identify top-performing or highly active hosts.

Section 3- Data Sources

StaySense BI primarily uses Airbnb listing data, including property details, pricing, room type, host information, neighbourhood, availability, and customer reviews. Additional datasets that could strengthen the platform include seasonal tourism trends, local events, weather information, competitor pricing, and demographic statistics. Combining internal listing data with external market data would provide richer insights, improve forecasting accuracy, and support better pricing and investment decisions.

Section 4- Recommended Tool Stack

The recommended BI stack consists of:

- **Google Forms or APIs** for data collection,
- **CSV files or PostgreSQL** for data storage,
- **Python (Pandas)** for data cleaning and analysis,
- **SQL** for querying datasets,
- **Power BI** for dashboard creation and visualization,
- **ChatGPT or Microsoft Copilot** for generating insights and summarizing trends.

This combination is cost-effective, beginner-friendly, and scalable for future enhancements such as predictive analytics and automated reporting.



Section 5- Phased Implementation Plan

- The first phase focuses on collecting and cleaning Airbnb listing data while establishing a structured database.
- The second phase involves developing SQL queries, performing exploratory data analysis in Python, and building interactive Power BI dashboards to monitor KPIs.
- The final phase introduces automation, external data integration, AI-assisted insights, and predictive analytics to improve forecasting, pricing optimization, and long-term business decision-making.